# Numerical Methods: Newton-Raphson Method

Questions 1–21

**Question 1.** Use the Newton Raphson Method with the value of $x_1$ given to find the next approximation, $x_2$, to find a root of each of the following equations, giving your answers to 2 decimal places:

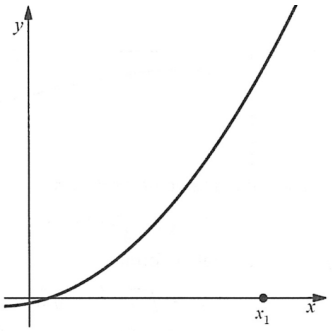
a) $x^4 + x^2 - 80 = 0$ using $x_1 = 3$
b) $x^3 + 2x^2 - 5x - 4 = 0$ using $x_1 = 2$
c) $e^x + x - 4 = 0$ using $x_1 = 1.5$

**Question 2.** Use the Newton Raphson Method with the value of $x_1$ given to find the next approximation, $x_2$, to find a solution of each of the following equations, giving your answers to 2 decimal places:

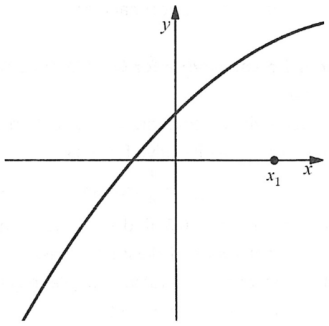
a) $2x^4 + x^3 = 5$ using $x_1 = 1.1$
b) $\frac{1}{6}x^6 - 8x = 4$ using $x_1 = 2$
c) $e^{2x} - x = 3$ using $x_1 = 0.65$

**Question 3.** For each graph below, draw a diagram to show the Newton-Raphson starting at the indicated point, labelling each of $x_2$, $x_3$ and $x_4$.

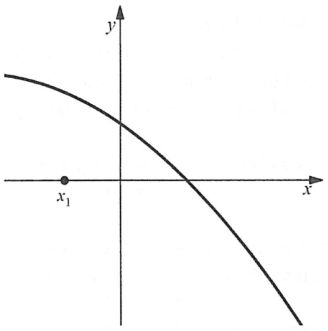
**a)**



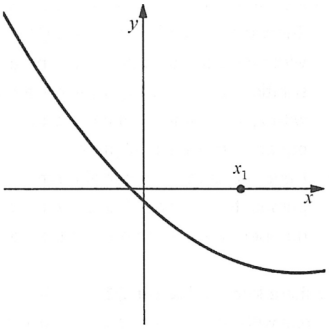

**b)**



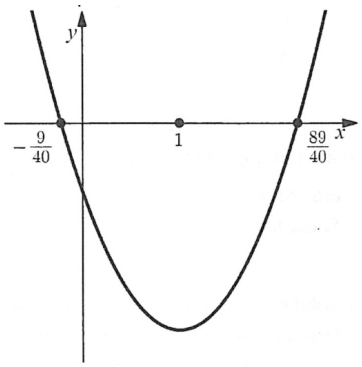

**c)**




**d)**



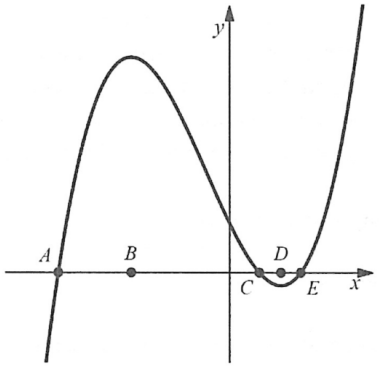

**Question 4.** The diagram shows $y = 2x^2 - 4x - 1$ with the $x$-values of the roots and the minimum point labelled on the $x$-axis.




Describe what happens when the Newton-Raphson iterative method is used with each of the following starting points:

a) $x_1 < 1$
b) $x_1 = 1$
c) $x_1 > 1$

**Question 5.** The diagram shows $y = x^3 + x^2 - 3x + 1$ with the $x$-values of the roots and the turning points labelled on the $x$-axis.




Describe what root is reached when using the Newton-Raphson method for each of the following starting points:

a) $x_1 = A$
b) $x_1 < B$
c) $x_1 = B$
d) $B < x_1 < D$
e) $x_1 = C$
f) $x_1 = D$
g) $x_1 > D$
h) $x_1 = E$

**Question 6.** By considering the root of the equation $x^3 - 20 = 0$ near $x = 3$, use Newton-Raphson method to find $\sqrt[3]{20}$ correct to 3 decimal places.

**Question 7.** The equation $3x^3 + 4x^2 - 3 = 0$ has one real root.

a) By considering a change in sign, show that there is a root between $x = 0$ and $x = 1$.

b) Show that the Newton-Raphson formula for finding the root can be written in the form:

$$x_{n+1} = \frac{6x_n^3 + 4x_n^2 + 3}{9x_n^2 + 8x_n}$$

c) Use the Newton-Raphson method with $x_1 = 1$ to find $x_2$ and $x_3$ to 4 decimal places.

d) Explain why the method fails when $x_1 = 0$ or $x_1 = -\frac{8}{9}$.

**Question 8. Prerequisite knowledge for Q8: Arithmetic Series** An arithmetic series has first term 2 and common difference 3. The sum of the first $n$ terms is $\frac{12}{n}$.

a) Show that $n$ satisfies the equation $3n^3 + n^2 - 24 = 0$.

b) Apply the Newton-Raphson method to the equation above using $n_1 = 2$ to obtain an answer correct to 4 significant figures.

**Question 9. Prerequisite knowledge for Q9: Geometric Series** A geometric series has first term 2 and common ratio $x$. The sum of the first 8 terms is 18.

a) Show that $x$ satisfies the equation $x^8 - 9x + 8 = 0$.

b) Apply the Newton-Raphson method to the equation above using $x_1 = 0.9$ to obtain an answer correct to 4 significant figures.

c) Hence determine the sum to infinity of the geometric series giving your answer to 3 significant figures.

**Question 10. Prerequisite knowledge for Q10: Modulus Functions**

a) Sketch the graph of $y = 5 - \frac{1}{x^2}$ and $y = 4 - |2x|$ on the same axes showing clearly the points of intersection.

b) Show that $\alpha$, the negative root of $5 - \frac{1}{x^2} = 4 - |2x|$ satisfies the equation $2x^3 - x^2 + 1 = 0$.

c) Hence show that $\alpha$ may be found using a Newton-Raphson iterative formula which may be written in the form

$$x_{n+1} = \frac{-4x_n^3 + x_n^2 + 1}{2x_n - 6x_n^2}$$

d) Use the formula starting with $x_1 = -1$ to find an approximation for $\alpha$ to 3 decimal places.

e) State the starting values at which the Newton-Raphson iterative formulae will fail.

**Question 11. Prerequisite knowledge for Q11–Q14: Differentiation of Standard Functions**

a) Show that $\sin x = 2 - x$ has a solution between $x = 1$ and $x = 2$.

b) Using the Newton-Raphson method and $x_1 = 0.4$ find the solution to $\sin x = 2 - x$ to 3 significant figures.

**Question 12.**

a) Draw a sketch of $y = \ln x$ and $y = 3 - x$ on the same axes for $0 \leq x \leq 3$.

b) By considering a suitable change in sign, show that the intersection of the two graphs lies between $x = 1$ and $x = 3$.

c) Use the Newton-Raphson method with $x_1 = 2$ to find the solution to 3 significant figures.

**Question 13.** The equation $x^2 + \ln(5x) = 0$ has one real root.

a) Show that there is a root between $x = 0.1$ and $x = 0.2$.

b) Show that the Newton-Raphson formula can be written in the form:

$$x_{n+1} = \frac{x_n^3 + x_n - x_n \ln(5x_n)}{2x_n^2 + 1}$$

c) Use the Newton-Raphson method starting with $x_1 = 0.1$ to find $x_2$, $x_3$ and $x_4$ to 4 decimal places.

d) Explain why the Newton-Raphson method can never fail to find the root.

**Question 14.** A curve has equation $y = 3x - x^2 + 10\ln(kx + 1)$, $x > 0$ where $k$ is a positive constant. It has a local maximum when $x = 3$.

a) Show that $k = 3$.

b) Show that the curve intersects the axes between $x = 7$ and $x = 8$.

c) Use the Newton-Raphson method with initial value $x_1 = 7$ to find the $x$-coordinate of this point of intersection to 3 decimal places.

**Question 15. Prerequisite knowledge for Q15–Q20: Product Rule**

a) Show that $x \sin x = 1$ has a solution between $x = 1$ and $x = 2$.

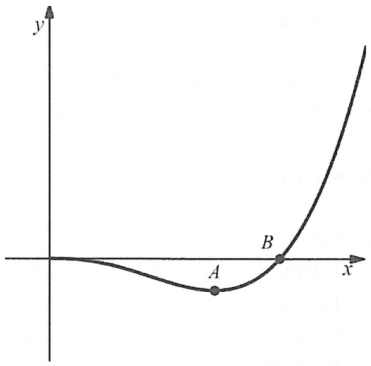
b) Using the Newton-Raphson method and $x_1 = 1$ find a solution to $x \sin x = 1$ giving your answer correct to 3 significant figures.

**Question 16.** Amy is trying to solve $2^x x^2 = 3$. She sets $f(x) = 2^x x^2 - 3$ and decides to use the Newton-Raphson method, starting at $x_1 = 0$.

a) Explain why she will not succeed in finding the solution.

b) Use the Newton-Raphson method at $x_1 = 1$ to find $x_3$, an approximation of the root giving your answer to 3 decimal places.

**Question 17.** The diagram below shows $y = x^3 \ln x$, $x > 0$ with the local minimum and the root labelled as $A$ and $B$ respectively.



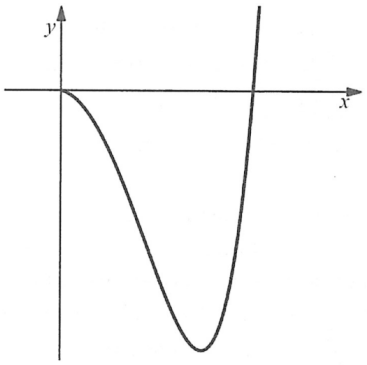

a) State the coordinates of point $B$.
b) Show that $\frac{d^2y}{dx^2} = 5x + 6x \ln x$.
c) By using Newton-Raphson's method and starting at $x_1 = 1$, show that the $x$-coordinate of the minimum point $A$ is 0.717 to 3 significant figures.

**Question 18.** The equation $x^2e^x - 3x + 1 = 0$ has two roots.
a) Show that there is a root between $x = 0.4$ and $x = 0.5$.
b) Show that the Newton-Raphson formula to find the roots of $x^2e^x - 3x + 1 = 0$ can be written in the form:

$$x_{n+1} = \frac{x_n^3 e^{x_n} + x_n^2 e^{x_n} - 1}{x_n^2 e^{x_n} + 2x_n e^{x_n} - 3}$$

c) Use the above iterative formula with starting point $x_1 = 0.4$ to find $x_2$ and $x_3$ giving your answers to 3 significant figures.
d) Use the Newton-Raphson method starting at $x_1 = 0.6$ to find the root of $x^2e^x + 2xe^x - 3 = 0$ giving your answer to 3 significant figures.
e) State the relevance of the result you found in part d) to with regards the curve $y = x^2e^x - 3x + 1$.

**Question 19.** The curve $y = x \times 3^x - 8x^2 - 2x$, $x > 0$ has a single stationary point which is close to $x = 2.2$.




a) Show that the $x$-coordinate of this stationary point satisfies the equation $3^x + (3^x \ln 3 - 16)x - 2 = 0$.
b) Show that the Newton-Raphson iterative formula for finding the $x$-value of the stationary point is

$$x_{n+1} = x_n - \frac{3^{x_n} + (3^{x_n} \ln 3 - 16)x_n - 2}{(2 \ln 3 + (\ln 3)^2 x_n)3^{x_n} - 16}$$

c) Find the $x$-coordinate of the stationary point giving it correct to 3 significant figures.

**Question 20.** The number of views per day of a video $N$ on a social media platform may be modelled by

$$N = 8e^{10-0.25t} \sin(3.15 - 0.19t)$$

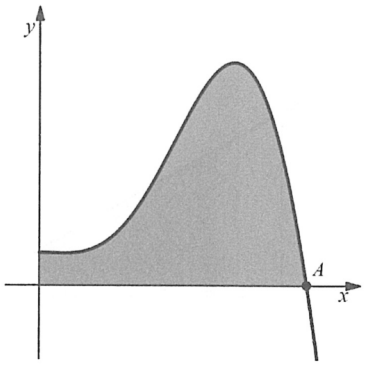
where $t$ is the number of days passed since the video was shared. The maximum of the model is found to occur when $t \approx 3.4646$.

a) Show that that the number of days passed when it reaches the maximum number of views per day is approximately 3.4646.
b) Use Newton-Raphson's method to find the number of days passed to the nearest hour when the views per day is zero.

Comment on the validity of the model

c) when 0 days have passed
d) after the number of days calculated in part b) has passed.

**Question 21. Prerequisite knowledge for Q21: Trapezium Rule** The diagram below shows $y = 0.1 + 0.1x^2 \sin(x - 0.5)$, $x > 0$ with the root labelled as $A$.




The shaded region is bounded by $x = 0$, $y = 0$ and where the graph meets the $x$-axis.

a) Show that the $x$-coordinate of the point $A$ lies between $x = 3$ and $x = 4$.

b) Use the Newton-Raphson method, to find an approximation for the $x$-coordinate of $A$ giving your answer to 3 decimal places.

c) Using your $x$-coordinate for $A$, use the trapezium rule with 4 strips to find an approximate value for the shaded area. Give your answer to 1 decimal place.